# A rotational sub-cell descriptor gives stable cross-topology prediction of effective stiffness where pixel surrogates do not

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**Abstract**

Machine-learned surrogates for the homogenized stiffness of periodic metamaterials are usually evaluated by splitting a pooled cell dataset at random. On a benchmark containing several unit-cell topology families this measures interpolation within known families, not the transfer a practitioner needs. We report three results on a purpose-built, duplicate-free 18-family benchmark. First, the protocol matters more than the model: the same surrogate scores $R^2$ between 0.90 and 0.96 under a random split and between $-0.73$ and $+0.14$ when whole topology families are held out, a gap of 0.58 to 1.63 across four stiffness components. Second, a three-number descriptor of how the softest sub-cell elastodynamic mode divides its motion between rotation, dilatation and shear – obtained from the same periodic operator that produces the stiffness tensor, at a quarter of the cost of one homogenization – is the only representation we tested that never fails: across twelve component-by-learner combinations its held-out $R^2$ stays within $[+0.02, +0.41]$, whereas 2304 raw pixels span $[-42.99, +0.34]$ and a published geometric spectral descriptor spans $[-1.56, +0.33]$. Third, its advantage is not uniform and we report where it fails: it beats the geometric spectrum and the free solid-fraction control on the shear-coupled components $C_{12}$ and $C_{22}$ for every learner tested, but does not beat solid fraction at all on $C_{11}$. That pattern is what a rotational-mode explanation predicts, and it is the reason we do not claim a general advantage. We also document a defect in the dataset the earlier version of this work used: rasterizing low-parameter shape families onto a fixed grid produces massive duplication, 61.6% in that case, which inflates in-distribution scores and depresses pixel baselines.

**Keywords:** computational homogenization; metamaterials; machine-learning surrogates; out-of-distribution generalization; micropolar effects; benchmark design

## 1 Introduction

Numerical homogenization returns an effective stiffness $\mathbf{C}^H$ for a periodic microstructure at the cost of one cell problem per cell. Because that cost is prohibitive inside a design loop, a large literature replaces it with a learned surrogate mapping geometry to $\mathbf{C}^H$, with reported accuracy of $R^2 \geq 0.98$ [1, 2].

That accuracy is measured within a design space. Whether it survives a change of unit-cell topology is a separate question, and a recent broad study of materials-property models concludes that most out-of-distribution evaluations in the field measure interpolation rather than extrapolation [3]. Work on generalizable microstructure–property maps across microstructure classes reaches a related conclusion and additionally reports that a distributional distance predicts when transfer will succeed [4].

This paper asks a narrower question with a controlled benchmark: when a surrogate is required to predict $\mathbf{C}^H$ for a topology family it has never seen, how much does the evaluation protocol matter, and is there a cheap physically-motivated representation that degrades gracefully rather than catastrophically?

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Our starting point is that $\mathbf{C}^H$ is obtained by volume-averaging the microscopic stress. Averaging is a projection: it retains the two rigid-translation modes of the cell and discards every other sub-cell degree of freedom. Roberts [5] constructs multi-continuum micromorphic homogenizations by retaining precisely those discarded modes, and the relaxed-micromorphic literature makes the same point constitutively — a first-order Cauchy continuum has no slot for the cell's rotational degree of freedom [6]. We use those modes not as a modelling target but as an inexpensive descriptor, and we test where that descriptor helps and where it does not.

**Contributions.** (i) MetaXFam-D, an 18-family, 3096-cell benchmark verified to contain no duplicate geometries, with family-disjoint evaluation (§2). (ii) A measurement of how much a random split overstates cross-topology performance (§3). (iii) A three-number rotational sub-cell descriptor that is the only representation tested whose held-out $R^2$ never becomes negative (§5). (iv) An explicit map of where it beats the baselines and where it does not, with confidence intervals and a pre-registered aggregation rule (§6). (v) A dataset-construction caution: rasterization collapse (§2.1).

**What we do not claim.** We do not claim the descriptor generally outperforms a pixel surrogate; on a duplicate-free benchmark it does not (§6). We do not claim non-injectivity of homogenization as novel; it is established [7]. We do not claim that distributional criteria fail to predict transfer — on this benchmark they partly succeed, consistent with [4].

# 2 Benchmark

Cells are $48 \times 48$ binary grids, plane strain, solid $E = 1$, $\nu = 0.3$, void $E = 10^{-6}$, with periodicity imposed by identifying opposite-edge nodes. $\mathbf{C}^H$ follows from volume-averaging element stresses under three unit macroscopic strains [12]; the solver passes six validation tests including the exact rank-1 laminate and Voigt–Reuss bracketing.

Every cell satisfies three conditions: solid fraction in $[0.45, 0.75]$; the solid phase percolates in both directions; and its rasterization is unique within its family. The benchmark comprises 18 families $\times$ 172 cells $=$ 3096 cells, verified to contain 3096 distinct geometries and no cross-family duplicates. Family median solid fractions span 0.579–0.695, so density is controlled and a cross-family error is a topology effect rather than a density extrapolation.

## 2.1 Rasterization collapse, and why the benchmark had to be rebuilt

An earlier version of this dataset sampled continuous shape parameters and kept 600 cells per family. Auditing it revealed that its 13,200 cells comprise only 5072 distinct rasterizations — a duplicate rate of 61.6%. Low-parameter families are the worst affected: one lattice family admits only *three* distinct $48 \times 48$ rasterizations inside the density band, and rejects essentially every further draw as already seen.

The mechanism is confirmed by resolution dependence (Fig. 1): duplication rises from 72.8% at $48 \times 48$ to 98.4% at $12 \times 12$. Coarser rasterization admits fewer distinct images. The consequence is severe: under a random $75/25$ split of that dataset, 67.7% of test cells have their exact geometry in the training set.

This is not a general property of pixelized metamaterial datasets. The published 2D subsets of METASET [9], audited with the same script, show 0% duplication at $50 \times 50$; that dataset selects for diversity explicitly and uses richer shape descriptions. The defect belongs to low-parameter parametric generators. We report it because the check is three lines of code and, in our case, it changed a headline conclusion.

## 2.2 Evaluation protocol

Training subsets of $K = 4$ families are drawn at random; 30 subsets $\times$ 2 model seeds gives 60 paired evaluations per arm, scored as pooled $R^2$ over the 14 held-out families. Three surrogate classes are

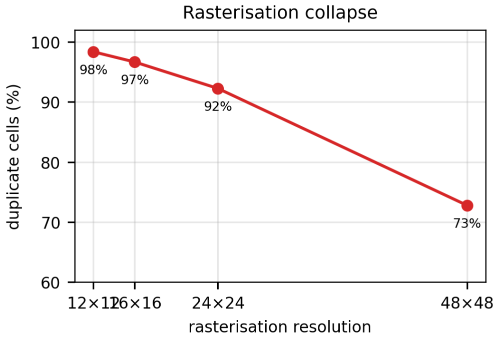


**Figure 1:** Duplication against rasterization resolution, measured on an eight-family probe. Distinct shape parameters collapse onto identical pixel images as resolution falls.

**Table I:** Held-out $R^2$ by protocol, ridge regression on raw pixels, median over 20 training subsets. Only the split protocol differs.

| component | random, same families | random, pooled | family-disjoint | gap |
|---|---|---|---|---|
| $C_{11}$ | +0.961 | +0.859 | +0.142 | 0.819 |
| $C_{12}$ | +0.897 | +0.745 | −0.728 | 1.625 |
| $C_{22}$ | +0.947 | +0.829 | −0.234 | 1.181 |
| $C_{33}$ | +0.912 | +0.752 | −0.416 | 1.328 |

used throughout — random forest, ridge regression with cross-validated regularization, and a two-layer perceptron — because representation comparisons that hold for one learner need not hold for another.

**Aggregation rule, fixed in advance.** A claim is reported as supported only if it holds for *every* learner on a given component, with a bootstrap 95% confidence interval on the paired median gain excluding zero. We state this because an earlier analysis of ours summarized on a single learner and reached a different headline from the same measurements.

## 3 The protocol gap

Table I compares three protocols at matched training-set size: a random split within the training families, a random split pooled over all families, and a family-disjoint split. The same model and the same number of training cells are used throughout; only the split changes.

The pooled random split — the standard practice when a dataset contains several families — still reports $R^2$ between 0.75 and 0.86 while the same surrogate is at or below zero on unseen topologies (Fig. 2). A practitioner following that protocol would report a strong result and ship a model that does not transfer.

## 4 The descriptor

On the same periodic operator, with a consistent mass matrix, we solve $\mathbf{K}\phi = \omega^2\mathbf{M}\phi$. Periodicity admits exactly two rigid modes; the modes above them are what stress-averaging discards. For each eigenvector the displacement gradient splits pointwise and exactly into dilatation, deviatoric shear and rotation,

$$|\nabla u|^2 = \tfrac{1}{2}d^2 + \tfrac{1}{2}\left[(\varepsilon_{xx} - \varepsilon_{yy})^2 + \gamma_{xy}^2\right] + 2w^2, \qquad w = \tfrac{1}{2}(\partial_x v - \partial_y u), \tag{1}$$

and each term is integrated over the cell weighted by element stiffness and normalized to sum to one. We call the result the *mode character* of that eigenvector, and use the character of the softest non-rigid mode: three numbers.

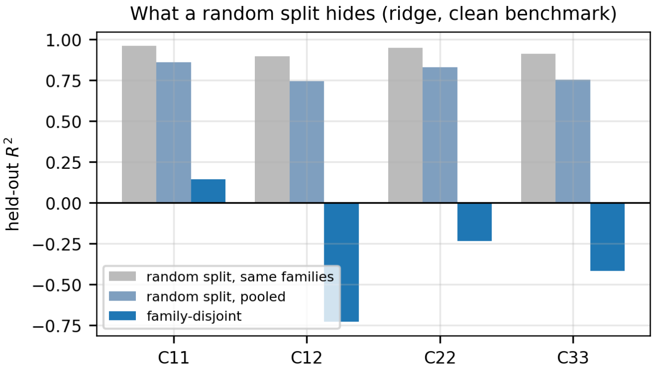


**Figure 2:** The same surrogate under three evaluation protocols on the clean benchmark.

**Table II:** Range of median held-out $R^2$ across four components $\times$ three learners.

| representation | dim | min | max |
|---|---|---|---|
| raw pixels | 2304 | $-42.99$ | $+0.34$ |
| solid fraction + shape-DNA | 7 | $-1.56$ | $+0.33$ |
| solid fraction | 1 | $+0.04$ | $+0.24$ |
| **solid fraction + mode-2 character** | **4** | **$+0.02$** | **$+0.41$** |

An eigenvector inside a degenerate cluster is defined only up to a rotation within the cluster and the character is quadratic in the eigenvector, so per-eigenvector character is basis-dependent. Every four-fold-symmetric family has a degenerate second mode. We therefore average over the cluster, which is basis-invariant, and use a deterministic eigensolver start.

**Validation.** On a homogeneous cell the exact periodic modes are plane waves: the lowest non-rigid cluster is the shear wave at $\omega = 2\pi\sqrt{\mu} = 3.8967$ with character exactly $(0, \frac{1}{2}, \frac{1}{2})$; the pressure wave has $(\frac{1}{2}, \frac{1}{2}, 0)$. The extractor reproduces the fundamental to $7.1 \times 10^{-4}$, converges as $h^2$, reproduces both exact characters to $10^{-6}$, and is invariant to a random rotation within a degenerate cluster to $10^{-14}$. Ten tests pass.

**Cost.** Computed on a $24 \times 24$ coarsened cell, the eigensolve costs $0.25\times$ one $48 \times 48$ homogenization.

## 5 Stability across surrogate classes

The clearest result concerns not average accuracy but the absence of failure. Table II reports, for each representation, the range of median held-out $R^2$ over all twelve component-by-learner combinations.

The pixel surrogate attains the single best cell in the table and also the worst by a factor of a hundred: under a perceptron it reaches $-42.99$ on $C_{33}$ and $-18.12$ on $C_{12}$. The four-number representation never becomes negative in any of the twelve cells (Fig. 3). Solid fraction alone is similarly stable but reaches only $+0.24$.

We regard this as the practically useful statement. A practitioner choosing a surrogate for an unseen topology cannot know in advance which learner will be well matched to the data; a representation that is mediocre-but-never-catastrophic is more valuable than one whose expected performance is higher but whose variance across model classes spans forty units of $R^2$.

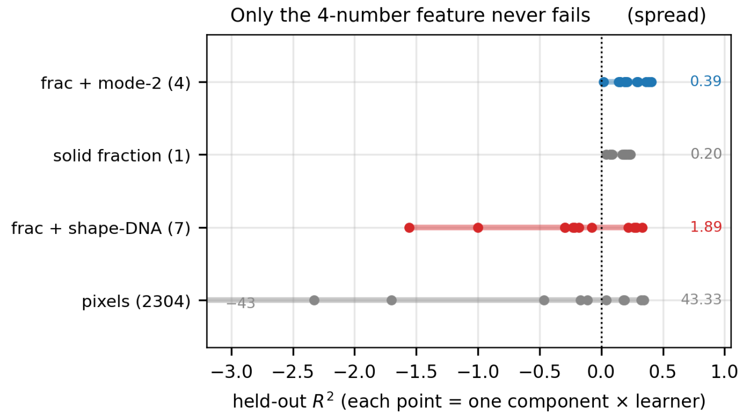


**Figure 3:** Each point is one component × learner combination.

## 6 Where the descriptor helps, and where it does not

Applying the pre-registered rule, none of the three pairwise claims holds across all four components (Fig. 4). The pattern is systematic.

**Against the free control (solid fraction).** The descriptor wins on $C_{22}$ for every learner (+0.315, +0.212, +0.195; all intervals clear of zero; 90–95% of paired evaluations) and on $C_{12}$ for two of three. On $C_{11}$ it wins for none: the gains are +0.046, −0.037, −0.060, with intervals containing zero.

**Against the published geometric spectrum.** Shape-DNA [8] is the Laplace–Beltrami spectrum of the cell — a spectrum, rotation invariant, purely geometric, and the closest prior art. The elastic descriptor beats it in ten of twelve cells, decisively on $C_{22}$ (+0.161 to +0.656, 90–100% of evaluations). It loses on $C_{11}$ under a random forest (−0.164, interval entirely negative). So it is not the case that any spectrum transfers: the elastic one does on the shear-coupled components and the geometric one largely does not.

**Against raw pixels.** The descriptor wins under ridge and the perceptron on every component, but under a random forest it is indistinguishable on $C_{22}$ and $C_{33}$ and loses on $C_{11}$ (−0.187). An earlier version of this work, computed on the duplicated dataset, reported a uniform +0.364 advantage; that advantage was an artifact of duplication depressing the pixel baseline.

**Reading the pattern.** The advantage is concentrated in $C_{12}$ and $C_{22}$ and absent in $C_{11}$. Under the rotational-mode account this is the expected signature rather than an inconvenience: the discarded degree of freedom is a rotation, and rotation couples to shear, so the components it should inform are the shear-coupled ones. A descriptor that improved every component equally would in fact be harder to explain.

## 7 Discussion

**What changed from the earlier version, and why.** A previous version of this work reported that a low-dimensional physical feature broadly outperformed a pixel surrogate, on a dataset later found to be 61.6% duplicated. Recomputing every number on duplicate-free data removed that conclusion: the pixel baseline recovers from −0.008 to +0.37 on $C_{22}$ once duplicates are gone. We report this because the direction of the bias is instructive — duplication does not merely inflate in-distribution scores, it *depresses* high-dimensional baselines on cross-family evaluation, and so systematically flatters low-dimensional competitors.

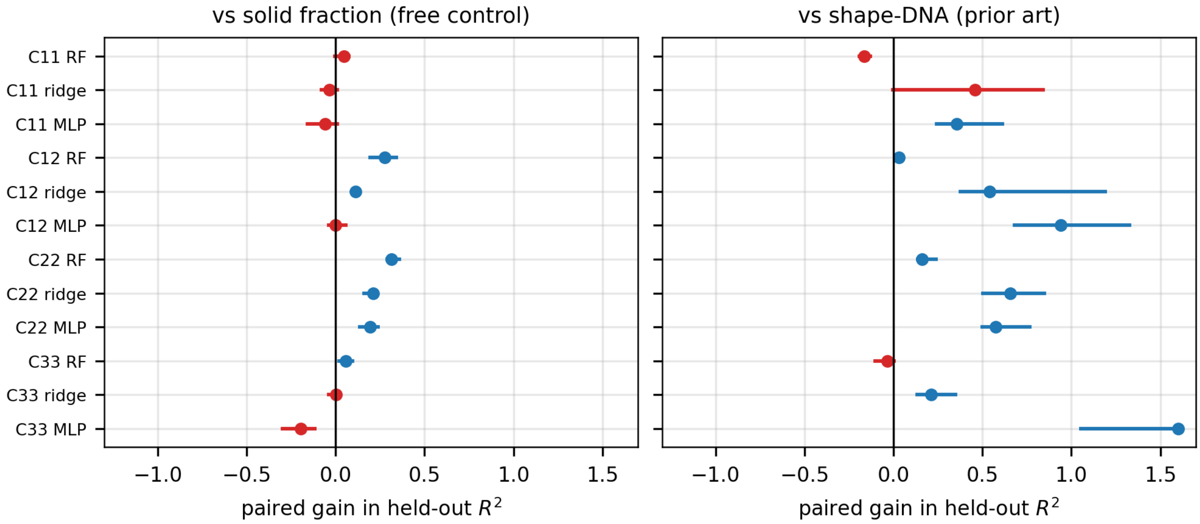


**Figure 4:** Paired gain of the four-number descriptor over each baseline, with bootstrap 95% intervals. Blue: interval excludes zero. Red: it does not. Values clipped for display.

**Relation to prior work.** Francis et al. [10] learn homogenized micropolar constants with a deep material network, extrapolating across constitutive behaviour and boundary conditions rather than topology. Hendriks et al. [11] address cross-topology generalization by making the architecture equivariant to geometric similarity; that is complementary to changing what is measured, and we do not compare against it here, which is the principal gap in this work. Liu et al. [4] report that a distributional distance predicts generalizability across microstructure classes; on this benchmark we reproduce that qualitatively.

**Limitations.** Two dimensions, linear elasticity, one contrast ratio, one void phase. The benchmark is 3096 cells; six families from the original pool could not supply enough distinct rasterizations and were re-parameterized, which changes what those families denote. No comparison against an equivariant graph network. The descriptor does not make cross-topology prediction accurate — the best held-out $R^2$ we report is $+0.41$.

## Data and code availability

The benchmark, generators, validated solver and duplication-audit tooling are archived at Zenodo, `https://doi.org/10.5281/zenodo.22167146`, and at `https://github.com/davidmashiah/metaxfam22`. The superseded duplicated dataset is retained at `https://doi.org/10.5281/zenodo.21734948` because published work refers to it. The sub-cell extractor and its validation suite are at `https://doi.org/10.5281/zenodo.22137052` and `https://github.com/davidmashiah/metaxfam-subcell`.

## Declaration of generative AI in the writing process

During the preparation of this work the author used a large language model as a coding and drafting assistant: to implement and run analysis scripts and to draft and revise prose. All numerical results were produced by the deposited code and checked by the author against the raw result files. All scientific claims, their framing, and the decision to report each negative result are the author's own. The author takes full responsibility for the content of this publication.

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